

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: POLYMERIZATION, RECOVERY, AND POWDER-FORM PROCESSING

1. ABSTRACT

This report synthesizes three primary patents relevant to nitrile butadiene rubber (NBR) preparation and post-polymerization handling, with emphasis on acrylonitrile/butadiene copolymerization and process control. The target material is low acrylonitrile NBR, defined here as NBR with reduced acrylonitrile incorporation relative to standard grades, typically sought to improve low-temperature flexibility while retaining oil and fuel resistance. Across the selected patents, the disclosed inventions span continuous emulsion polymerization, latex coagulation/powder recovery, and stabilized latex recovery with pH-controlled reslurrying. The landscape shows that NBR is predominantly produced by aqueous emulsion polymerization of acrylonitrile and 1,3-butadiene, with water serving as the polymerization medium in all three patents, though the downstream objectives differ substantially.

US5770660A focuses on a continuous, multi-vessel emulsion polymerization strategy in which butadiene is incrementally added to maintain a high acrylonitrile/butadiene monomer ratio and achieve a latex with about fifty percent bound acrylonitrile. The patent emphasizes composition control through staged butadiene feed, steady-state conversion management, and recovery operations including coagulation, washing, dewatering, drying, and baling. EP0071240A1 is centered on direct production of NBR powder from latex by using sodium lauryl sulfate as emulsifier and a critical magnesium/aluminum sulfate coagulant system; it demonstrates that powder-size coagulum can be obtained from NBR latices with bound acrylonitrile contents around 31–34.6%. US5708132A does not disclose NBR synthesis conditions, but it is relevant to NBR manufacturing because it improves recovery of nitrile rubber latex through antioxidant stabilization, salt/acid coagulation, wash-water reslurrying, and pH adjustment to reduce cure variability and improve storage stability. Collectively, the patents show that low-to-mid acrylonitrile NBR is typically made by aqueous emulsion copolymerization, while downstream water-based coagulation and washing steps are critical to product form, stability, and processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US5770660A
- Patent Title: Continuous polymerization process for producing NBR rubber ...
- Assignee: Individual
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a continuous free-radical emulsion copolymerization of acrylonitrile and butadiene in a series of reaction vessels. Acrylonitrile is charged with an initial fraction of butadiene, and additional butadiene is added incrementally in downstream vessels to maintain the desired monomer ratio and produce NBR latex with high bound acrylonitrile. - Acrylonitrile to butadiene charge ratio: about 80 parts acrylonitrile to 20 parts $\pm 10\%$ butadiene — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene — Claim 1 - Desired bound acrylonitrile content: about fifty percent by weight — Claim 1 - Desired acrylonitrile content: about forty-five percent to about fifty-five percent by weight — Detailed description - Example composition: 75 percent by weight acrylonitrile and 25 percent by weight butadiene — Detailed description - First vessel conversion: 24 percent — Example VII - Second vessel conversion: 40 percent — Example VII - Third vessel conversion: 56 percent — Example VII - Fourth vessel conversion: 72 percent — Example VII - Fifth vessel conversion: 82 percent — Example VII - Latex bound acrylonitrile content: 48.8 percent — Example VII - Latex bound acrylonitrile content: 59.5 percent — Example I - Latex bound acrylonitrile content: 52.6 percent — Example II - Latex bound acrylonitrile content: 54.9 percent — Example III - Latex bound acrylonitrile content: 50.5 percent — Example IV - Latex bound acrylonitrile content: 50.2 percent — Example V - Latex bound acrylonitrile content: 50.6 percent — Example VI - Medium & Water Role: Water is part of the emulsion polymerization medium used to form NBR latex, and later water is also used in recovery operations including coagulation, washing, and dewatering/drying of the latex-derived polymer. - acrylonitrile content: about fifty percent by weight bound acrylonitrile; preferred compositions about forty-five percent to about fifty-five percent by weight acrylonitrile

Medium & Water Role - Water is part of the emulsion polymerization medium used to form NBR latex, and later water is also used in recovery operations including coagulation, washing, and dewatering/drying of the latex-derived polymer. - Core reaction medium: aqueous emulsion / latex polymerization

acrylonitrile content - about fifty percent by weight bound acrylonitrile; preferred compositions about forty-five percent to about fifty-five percent by weight acrylonitrile

Relevant Experimental Evidence - Example I: Conventional batch addition produced 59.5% bound acrylonitrile but with low conversion and widely varying monomer ratio. - Example II:

Two-increment batch addition improved conversion to 73.3% and gave 52.6% bound acrylonitrile. - Example IV: Three butadiene increments gave 50.5% bound acrylonitrile at 85.2% conversion, showing improved control. - Example V: Four-increment batch feed gave 50.2% bound acrylonitrile at 81.9% conversion. - Example VII: Continuous multi-vessel feed produced 48.8% bound acrylonitrile and improved tensile/environmental resistance properties.

Technical Relevance This patent is highly relevant because it directly addresses NBR composition control through staged monomer feeding, a key lever for managing acrylonitrile incorporation. Although the disclosed compositions are not low-acrylonitrile grades, the process principles are directly applicable to controlling ACN content in NBR synthesis.

Patent 2

- Patent Number: EP0071240A1
- Patent Title: Production of emulsion-polymerized butadiene rubber in powder form
- Assignee: BF Goodrich Corp
- Jurisdiction: EP
- Publication Year: 1983
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses production of NBR latex by conventional aqueous emulsion polymerization of butadiene and acrylonitrile, followed by direct coagulation to obtain powder-form rubber. The key process feature is the use of sodium lauryl sulfate or related alkyl sulfate emulsifiers in the latex and a combined magnesium sulfate/aluminum sulfate aqueous coagulant to generate powder-size coagulum. - Latex total solids content: 34.6% — Example 1 - Latex pH: 8.0 — Example 1 - Bound acrylonitrile content: 34.6% — Example 1 - Mooney viscosity (ML-4-212): 61 — Example 1 - Latex amount: 100 parts — Example 1 - Aqueous coagulant solution: 870 parts — Example 1 - Magnesium sulfate: 4.4 phr — Example 1 - Aluminum sulfate: 2.2 phr — Example 1 - Coagulation temperature: 60°C — Example 1 - Latex total solids content: 32.6% — Examples 2 to 11 - Latex pH: 9.7 — Examples 2 to 11 - Bound acrylonitrile content: 31% — Examples 2 to 11 - Mooney viscosity (ML-4-212): 55 — Examples 2 to 11 - Combined coagulants: 0.05 to 0.09 ephr — Examples 2 to 11 - Mg/Al molar ratio: 0.3 to 1 to 1 to 1 — Examples 2 to 11 - Coagulation temperature: 60-80°C — Examples 2 to 11 - Magnesium sulfate: 25 parts — Examples 12 to 16 - Aluminum sulfate: 18 parts — Examples 12 to 16 - Water: 7500 parts at 80°C — Examples 12 to 16 - Water: 7500 parts of 80°C water — Examples 12 to 16 - Drying air: 220°F — Examples 12 to 16 - Zinc soap: 1 to 10 phr — Isolation of Dry Powdered Rubber - Zinc stearate: 3 phr — Examples 17 to 20 - Medium & Water Role: Water is explicitly part of the emulsion polymerization medium that forms the NBR latex, and it is also used later for aqueous coagulation, washing, and wet-cake handling before drying to powder. - acrylonitrile content: 34.6% bound acrylonitrile; 31% bound acrylonitrile in similar latexes

Medium & Water Role - Water is explicitly part of the emulsion polymerization medium that forms the NBR latex, and it is also used later for aqueous coagulation, washing, and wet-cake handling before drying to powder. - Core reaction medium: aqueous emulsion / latex polymerization

acrylonitrile content - 34.6% bound acrylonitrile; 31% bound acrylonitrile in similar latexes

Relevant Experimental Evidence - Example 1: Sodium lauryl sulfate latex coagulated with $\text{MgSO}_4/\text{Al}_2(\text{SO}_4)_3$ produced fine powder-size rubber particles. - Examples 2 to 11: Varying coagulant amounts and procedures showed best powder formation with 0.05 to 0.09 ephr combined coagulants and Mg/Al ratios of 0.3:1 to 1:1. - Examples 12 to 16: Washing with 80°C water and drying at 220°F produced powder-size coagulum; zinc stearate improved anti-agglomeration. - Examples 17 to 20: Zinc soap anticake additives yielded dry free-flowing powder suitable for rubber compounding.

Technical Relevance This patent is relevant because it discloses an NBR latex route and demonstrates how emulsifier and coagulation chemistry control the physical form of the recovered rubber. The bound acrylonitrile levels reported are in the low-to-mid NBR range and the process is directly applicable to manufacturing low-acrylonitrile NBR powders.

Patent 3

- Patent Number: US5708132A
- Patent Title: US5708132A - Method for the production of nitrile rubber
- Assignee: Goodyear Tire and Rubber Co
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent does not disclose a new NBR polymerization route; instead, it improves recovery of nitrile rubber from an already prepared aqueous emulsion latex. The process adds antioxidant to the latex, coagulates with salt and acid, separates crumb, reslurries the crumb in wash water, adjusts pH to about neutral, and then separates and optionally dries the rubber. - Antioxidant addition: 1.5 percent tris(nonylphenol) phosphite — Comparative Example 1 - Unhydrolyzed tris(nonylphenol) phosphite antioxidant: 0.04 percent — Comparative Example 1 - Hydrolyzed tris(nonylphenol) phosphite antioxidant: 0.54 percent — Comparative Example 1 - Unhydrolyzed tris(nonylphenol) phosphite antioxidant: 0.01 percent — Comparative Example 2 - Hydrolyzed tris(nonylphenol) phosphite antioxidant: 0.77 percent — Comparative Example 2 - Unhydrolyzed tris(nonylphenol) phosphite antioxidant: 0.02 percent — Comparative Example 3 - Hydrolyzed tris(nonylphenol) phosphite antioxidant: 2.20 percent — Comparative Example 3 - Antioxidant addition: 0.8 percent tris(nonylphenol) phosphite — Example 4 - Antioxidant addition: 0.2 percent Ultrinox™ 276 — Example 4 - Unhydrolyzed tris(nonylphenol) phosphite antioxidant: 0.31 percent — Example 4 - Hydrolyzed tris(nonylphenol)

phosphite antioxidant: 0.71 percent — Example 4 - Unhydrolyzed tris(nonylphenol) phosphite antioxidant: 0.37 percent — Example 5 - Hydrolyzed tris(nonylphenol) phosphite antioxidant: 0.49 percent — Example 5 - Antioxidant addition: 0.2 percent Ultrinox™ 276 — Examples 6 and Comparative Example 7 - Aging temperature: 70° C — Examples 6 and Comparative Example 7 - Aging time: up to 28 days — Examples 6 and Comparative Example 7 - Antioxidant addition: 2 percent tris(nonylphenyl) phosphite — Examples 8-12 and Comparative Example 13 - Compounding zinc oxide: 5 phr — Examples 8-12 and Comparative Example 13 - Compounding Captax® 2-mercaptobenzothiazole accelerator: 1 phr — Examples 8-12 and Comparative Example 13 - Compounding methyl zimate: 0.5 phr — Examples 8-12 and Comparative Example 13 - Compounding sulfur: 2 phr — Examples 8-12 and Comparative Example 13 - Rheometer temperature: 280° F. (138° C.) — Examples 8-12 and Comparative Example 13 - Example 8 pH: 3.0 — Table II - Example 9 pH: 4.5 — Table II - Example 10 pH: 6.0 — Table II - Example 11 pH: 7.0 — Table II - Example 12 pH: 8.0 — Table II - Comparative Example 13 pH: not adjusted — Table II - TC50: 5.8 min — Example 8 - TC90: 7.8 min — Example 8 - TC50: 5.6 min — Example 9 - TC90: 7.7 min — Example 9 - TC50: 5.3 min — Example 10 - TC90: 7.5 min — Example 10 - TC50: 4.9 min — Example 11 - TC90: 7.3 min — Example 11 - TC50: 5.2 min — Example 12 - TC90: 7.4 min — Example 12 - TC50: 7.2 min — Comparative Example 13 - TC90: 8.8 min — Comparative Example 13 - Medium & Water Role: The nitrile rubber is handled as an aqueous latex, then coagulated and reslurried in wash water before pH adjustment and separation; water is therefore central to the recovery and post-treatment sequence rather than the polymerization step itself in the extracted examples. - acrylonitrile content: Not disclosed in extracted evidence

Medium & Water Role - The nitrile rubber is handled as an aqueous latex, then coagulated and reslurried in wash water before pH adjustment and separation; water is therefore central to the recovery and post-treatment sequence rather than the polymerization step itself in the extracted examples. - Core reaction medium: aqueous emulsion / latex recovery process

acrylonitrile content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Example 1: Latex antioxidant addition before coagulation left low residual unhydrolyzed antioxidant in the dry rubber. - Example 4: Adding Ultrinox™ 276 with tris(nonylphenol) phosphite greatly increased retained unhydrolyzed antioxidant after recovery. - Example 6: pH adjustment of the reslurry influenced aging behavior of the recovered nitrile rubber. - Examples 8 to 12: Reslurry pH adjustment reduced cure times, with best results near pH 7.0. - Comparative Example 13: No pH adjustment gave slower cure times than the adjusted examples.

Technical Relevance This patent is relevant to low-acrylonitrile NBR manufacturing because it addresses recovery and stabilization of nitrile rubber latex, which is a critical downstream step after emulsion polymerization. Although it does not disclose the polymer composition, it is directly applicable to NBR latex handling and product quality control.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
US5770660A	Individual	1998	Water is part of the emulsion polymerization medium used to form NBR latex, and later water is also used in recovery operations including coagulation, washing, and dewatering/drying of the latex-derived polymer.	about fifty percent by weight bound acrylonitrile; preferred compositions about forty-five percent to about fifty-five percent by weight acrylonitrile	Example I: Conventional batch addition produced 59.5% bound acrylonitrile but with low conversion and widely varying mon

EP0071240A1	BF Goodrich Corp	1983	Water is explicitly part of the emulsion polymerization medium that forms the NBR latex, and it is also used later for aqueous coagulation, washing, and wet-cake handling before drying to powder.	34.6% bound acrylonitrile; 31% bound acrylonitrile in similar latexes	Example 1: Sodium lauryl sulfate latex coagulated with $\text{MgSO}_4/\text{Al}_2(\text{SO}_4)_3$ produced fine powder-size rubber particles.
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US5708132A	Goodyear Tire and Rubber Co	1998	The nitrile rubber is handled as an aqueous latex, then coagulated and reslurried in wash water before pH adjustment and separation; water is therefore central to the recovery and post-treatment sequence rather than the polymerization step itself in the extracted examples.	Not disclosed in extracted evidence	Comparative Example 1: Latex antioxidant addition before coagulation left low residual unhydrolyzed antioxidant in the d
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5770660A and EP0071240A1 both use aqueous emulsion/latex NBR chemistry, but US5770660A is a continuous polymerization/composition-control patent while EP0071240A1 is a latex coagulation and powder-recovery patent.
- US5770660A discloses much higher target acrylonitrile incorporation, centered around about 50 wt% bound acrylonitrile, whereas EP0071240A1 reports NBR latices at 31-34.6 wt% bound acrylonitrile.

- US5770660A controls composition by staged butadiene addition across multiple reactors; EP0071240A1 controls product form by sodium lauryl sulfate emulsifier selection and Mg/Al sulfate coagulation chemistry.
- US5708132A is not a synthesis patent but a recovery/stabilization patent; it uses coagulation, wash-water reslurrying, and pH adjustment to improve the properties of recovered nitrile rubber.
- All three patents rely on water-based processing, but only US5770660A and EP0071240A1 explicitly disclose polymerization in aqueous emulsion; US5708132A focuses on post-polymerization latex recovery and washing.

4. CONCLUSION

The selected patents show that NBR manufacturing is predominantly aqueous-emulsion based, with composition control and recovery chemistry being the main levers for product quality. US5770660A provides the strongest disclosure on monomer-feed control for high-bound-acrylonitrile NBR, EP0071240A1 demonstrates direct powder recovery from NBR latex with low-to-mid acrylonitrile content, and US5708132A improves latex recovery and post-coagulation stability through antioxidant and pH management. Together, they define a practical process landscape for NBR synthesis and handling, even though only US5770660A directly targets composition control at the polymerization stage.

5. REFERENCES

- US5770660A | Continuous polymerization process for producing NBR rubber ... | Individual | US | 1998 | <https://patents.google.com/patent/US5770660A/en>
- EP0071240A1 | Production of emulsion-polymerized butadiene rubber in powder form | BF Goodrich Corp | EP | 1983 | <https://patents.google.com/patent/EP0071240A1/en>
- US5708132A | US5708132A - Method for the production of nitrile rubber | Goodyear Tire and Rubber Co | US | 1998 | <https://patents.google.com/patent/US5708132A/en>